

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

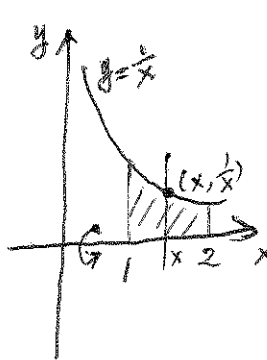
Problem 1. (5 points) Find the average value of the function $f(x) = \sqrt[3]{x}$ on $[1, 8]$.

Solution:

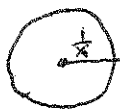
$$\begin{aligned}
 f_{avg} &= \frac{1}{8-1} \int_1^8 \sqrt[3]{x} \, dx \\
 &= \frac{1}{7} \cdot \frac{3}{4} x^{4/3} \Big|_1^8 \\
 &= \frac{3}{28} (8^{4/3} - 1^{4/3}) \\
 &= \frac{45}{28}
 \end{aligned}$$

Problem 2. (5 points) Find the volume of the solid obtained by rotating the region bounded by $y = 1/x$, $y = 0$, $x = 1$, $x = 2$ around the x -axis.

Solution:



$$\begin{aligned}
 V &= \int_1^2 \pi \cdot \left(\frac{1}{x}\right)^2 \, dx \\
 &= \int_1^2 \frac{\pi}{x^2} \, dx \\
 &= \pi \left(-\frac{1}{x}\right) \Big|_1^2 \\
 &= \pi \left[\left(-\frac{1}{2}\right) - (-1) \right] \\
 &= \frac{\pi}{2}
 \end{aligned}$$



$$A(x) = \pi \left(\frac{1}{x}\right)^2$$